# Alignment

**School of Electrical and Computer Engineering**

Ali Bavafa | November 12, 2026



# About me

- Research interests
  - Causality
  - Generative AI
  - Medical Image Analysis
- Hobbies
  - Movies
  - Books
  - Football, Chess



# Motivation

## Three H's of model desiderata



- LLMs need three properties to be applicable in real-world (A.K.A 3H's) [1]
  - **Helpfulness**: help the user solve their task according to the instructions
  - **Honesty**: give accurate information, express uncertainty when doesn't know the answer, instead of hallucinating
  - **Harmlessness**: not cause physical, psychological, or social harm to people or the environment.
- ***BUT...!***

[1] Askell, Amanda, et al. "A general language assistant as a laboratory for alignment." arXiv preprint arXiv:2112.00861 (2021).

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$$p(x) = \prod_{i=1}^{n} p(s_n | s_1, \cdots, s_n - 1)$$

**Training:** Next Token Prediction

$\not\Rightarrow$



The three H's of Model Desiderata

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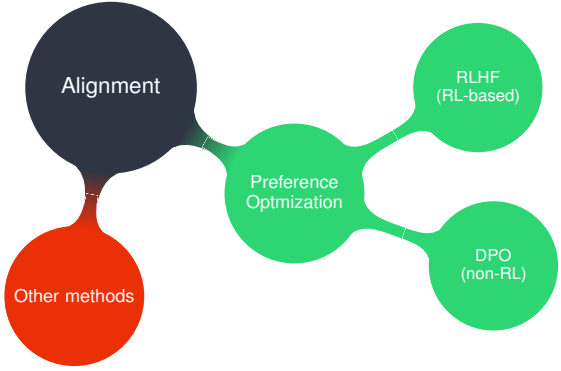

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# Motivation

## Alignment

**Alignment**

Follow the user's instructions helpfully, accurately and safely




Highlight:

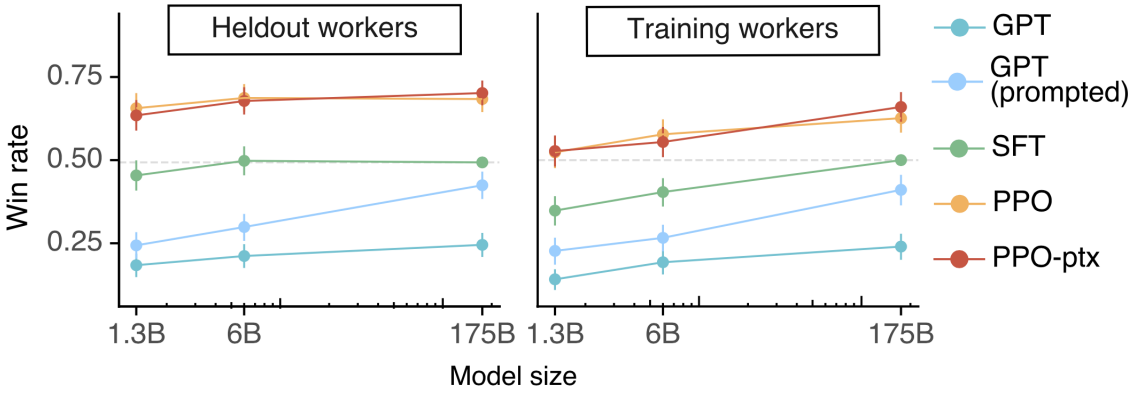
- Treat the model as a policy
- Use a reward signal
- Optimize through interaction and feedback

Highlight:

- No reward model
- No RL algorithm
- Directly fine-tune on preferences

# RLHF

## Impact




Ouyang, Long, et al. "Training language models to follow instructions with human feedback." NeurIPS 35 (2022)

# RLHF

**Reinforcement Learning with Human Feedback**[1]

- *RLHF* has 3 main steps
  1. Fine-tuning a pre-trained LM with supervised learning on high-quality data for the downstream task of interest (summarization, etc.), to obtain a policy model $\pi_{SFT}$
  2. Reward Modelling Phase

$$p^*\left(y_1 \succ y_2 \mid x\right)=\frac{\exp\left(r^*\left(x, y_1\right)\right)}{\exp\left(r^*\left(x, y_1\right)\right)+\exp\left(r^*\left(x, y_2\right)\right)} \tag{1}$$

$$\mathcal{L}_R\left(r_\phi, \mathcal{D}\right)=-\mathbb{E}_{\left(x, y_w, y_l\right) \sim \mathcal{D}}\left[\log \sigma\left(r_\phi\left(x, y_w\right)-r_\phi\left(x, y_l\right)\right)\right] \tag{2}$$

  3. RL Fine-Tuning Phase

$$\max_{\pi_\theta} \mathbb{E}_{x \sim \mathcal{D}, y \sim \pi_\theta(y \mid x)}\left[r_\phi(x, y)\right]-\beta \mathbb{D}_{\mathrm{KL}}\left[\pi_\theta(y \mid x) \| \pi_{\mathrm{ref}}(y \mid x)\right] \tag{3}$$

- maximize reward $r(x, y)=r_\phi(x, y)-\beta\left(\log \pi_\theta(y \mid x)-\log \pi_{\mathrm{ref}}(y \mid x)\right)$ using PPO

[1] Ouyang, Long, et al. "Training language models to follow instructions with human feedback." NeurIPS 35 (2022)

# DPO to the rescue!

## Direct Preference Optimization (DPO) [1]

- Implicit reward

$$\pi_r(y \mid x) = \frac{1}{Z(x)} \pi_{\text{ref}}(y \mid x) \exp\left(\frac{1}{\beta} r(x, y)\right),$$

$$\Rightarrow r(x, y) = \beta \log \frac{\pi_r(y \mid x)}{\pi_{\text{ref}}(y \mid x)} + \beta \log Z(x),\ Z(x) = \sum_y \pi_{\text{ref}}(y \mid x) \exp\left(\frac{1}{\beta} r(x, y)\right) \tag{4}$$

$$\mathcal{L}_{\text{DPO}}\left(\pi_\theta; \pi_{\text{ref}}\right) = -\mathbb{E}_{(x, y_w, y_l) \sim \mathcal{D}}\left[\log \sigma\left(\beta \log \frac{\pi_\theta\left(y_w \mid x\right)}{\pi_{\text{ref}}\left(y_w \mid x\right)} - \beta \log \frac{\pi_\theta\left(y_l \mid x\right)}{\pi_{\text{ref}}\left(y_l \mid x\right)}\right)\right] \tag{5}$$

- DPO update

$$\nabla_\theta \mathcal{L}_{\text{DPO}}\left(\pi_\theta; \pi_{\text{ref}}\right) = -\beta \mathbb{E}_{(x, y_w, y_l) \sim \mathcal{D}}[\underbrace{\sigma\left(\hat{r}_\theta\left(x, y_l\right) - \hat{r}_\theta\left(x, y_w\right)\right)}_{\text{higher weight when reward estimate is wrong}} [\underbrace{\nabla_\theta \log \pi\left(y_w \mid x\right)}_{\text{increase likelihood of } y_w} - \underbrace{\nabla_\theta \log \pi\left(y_l \mid x\right)}_{\text{decrease likelihood of } y_l}]]$$

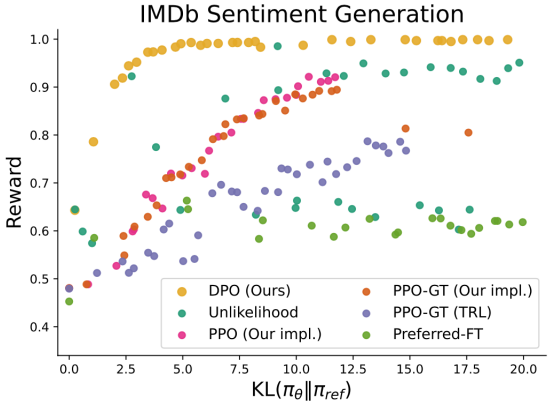
- About $\pi_{ref}$

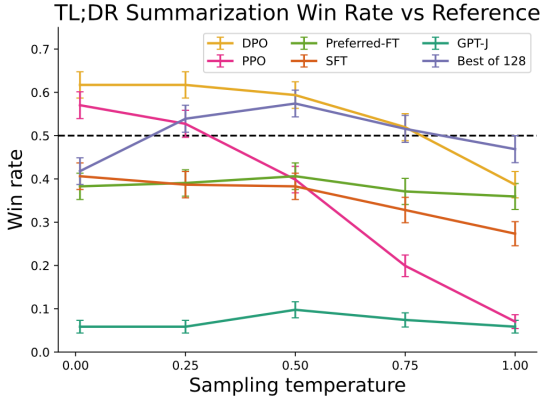
$$\pi_{\text{ref}} = \arg\max_\pi \mathbb{E}_{x, y_w \sim \mathcal{D}}\left[\log \pi\left(y_w \mid x\right)\right]. \tag{6}$$

[1] Rafailov, Rafael, et al. "Direct preference optimization: Your language model is secretly a reward model." NeurIPS 36 (2023)

# DPO

## Results




Rafailov, Rafael, et al. "Direct preference optimization: Your language model is secretly a reward model." NeurIPS 36 (2023)

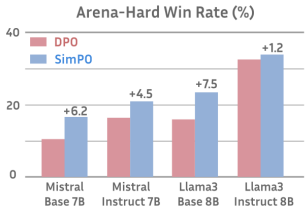
# Beyond DPO

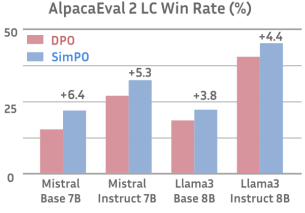
## SimPO[1]

- Challenges with DPO
  - Reward formulation is not directly aligned with generation metric
  - Ranking rewards ($r(x, y_w) > r(x, y_l)$) not necesserily mean ranking likelihoods ($p_\theta(y_w|x) > p_\theta(y_l|x)$)
  - DPO need $\pi_{ref}$

$\Rightarrow$ Proposed reward in SimPO:

$$r_{SimPO}(x, y) = \frac{\beta}{|y|} log\pi_\theta(y|x) = \frac{\beta}{|y|} \sum_{i=1}^{|y|} log\pi_\theta(y_i|x, y_{<i})$$




[1] Meng, Yu, Mengzhou Xia, and Danqi Chen. "Simpo: Simple preference optimization with a reference-free reward." NeurIPS (2024)

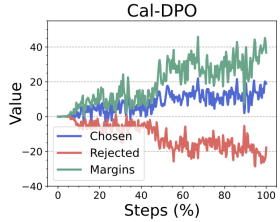
# Beyond DPO

## Cal-DPO[1]

- Challenges with DPO
  - Maximize the expected relative difference between the implicit rewards associated with the chosen and rejected responses
  - Ranking loss tend to ignore the absolute values of the rewards
  - May decrease log-likelihood of $y_w$

$\Rightarrow$ Calibrate preference data relative to the ground-truth rewards (scalewise): $\frac{\pi_\theta(\mathbf{y}|\mathbf{x})}{\pi_{\text{ref}}(\mathbf{y}|\mathbf{x})} = \frac{r(\mathbf{x},\mathbf{y})}{\beta}$

$$\mathcal{L}_{\text{Cal-DPO}}(\theta, \mathbf{x}, \mathbf{y_w}, \mathbf{y_l}) = -\log\sigma\left(\log\frac{\pi_\theta(\mathbf{y_w} \mid \mathbf{x})}{\pi_{\text{ref}}(\mathbf{y_w} \mid \mathbf{x})} - \log\frac{\pi_\theta(\mathbf{y_l} \mid \mathbf{x})}{\pi_{\text{ref}}(\mathbf{y_l} \mid \mathbf{x})}\right) + \left(\log\frac{\pi_\theta(\mathbf{y_w} \mid \mathbf{x})}{\pi_{\text{ref}}(\mathbf{y_w} \mid \mathbf{x})} - \frac{1}{2\beta}\right)^2 + \left(\log\frac{\pi_\theta(\mathbf{y_l} \mid \mathbf{x})}{\pi_{\text{ref}}(\mathbf{y_l} \mid \mathbf{x})} + \frac{1}{2\beta}\right)^2$$




[1] Xiao, Teng, et al. "Cal-DPO: Calibrated Direct Preference Optimization for Language Model Alignment." NeurIPS 38 (2024)

# Beyond DPO

## Causal-DPO[1]

- Motivation
  - Spurious correlations in reward modeling
  - Can lead to problems like reward hacking

$\Rightarrow$ Counterfactual invariance: $T(r(Z = z)) = T(r(Z = z')) \Rightarrow f(T) \perp\!\!\!\perp Z$

$\Rightarrow$ Maximum Mean Discrepancy (MMD) as regulizer to enforce independence

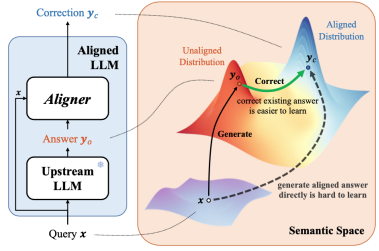
$$\mathrm{MMD}(\mathcal{F}, P, Q) = \sup_{f \in \mathcal{F}} \left(\mathbb{E}_{x \sim P}[f(x)] - \mathbb{E}_{y \sim Q}[f(y)]\right)$$

$$\mathcal{L}_{\mathrm{Causal-DPO}}\left(\pi_\theta; \pi_{\mathrm{ref}}\right) = -\mathbb{E}_{(x, y_w, y_l) \sim \mathcal{D}}\left[\log \sigma\left(\beta \log \frac{\pi_\theta\left(y_w \mid x\right)}{\pi_{\mathrm{ref}}\left(y_w \mid x\right)} - \beta \log \frac{\pi_\theta\left(y_l \mid x\right)}{\pi_{\mathrm{ref}}\left(y_l \mid x\right)}\right)\right] + \lambda \sum_{m, m' \in [M]} MMD\left(p_m(r(\mathbf{x}, \mathbf{y})), p'_m(r(\mathbf{x}, \mathbf{y}))\right)$$

[1] Wang, Chaoqi, et al. "Beyond Reward Hacking: Causal Rewards for Large Language Model Alignment." arXiv 2025

# Even more papers!

- Papers worth mentioning
  - Aligner[1], IPO[2], ORPO[3], KTO[4]
- Alignment for generative models of other modalities
  - SpeechAlign[5]
  - Diffusion-DPO[6]
- And more ...
- What's next?




Aligner (NeurIPS 2024)

[1] Ji, Jiaming, et al. "Aligner: Efficient alignment by learning to correct." NeurIPS 2024
[2] Azar, Mohammad Gheshlaghi, et al. "A general theoretical paradigm to understand learning from human preferences." AISTATS 2024
[3] Hong, Jiwoo, Noah Lee, and James Thorne. "ORPO: Monolithic Preference Optimization without Reference Model." EMNLP 2024
[4] Ethayarajh, Kawin, et al. "KTO: Model alignment as prospect theoretic optimization." arXiv 2024
[5] Zhang, Dong, et al. "SpeechAlign: Aligning Speech Generation to Human Preferences." NeurIPS 2024.
[6] Wallace, Bram, et al. "Diffusion model alignment using direct preference optimization." CVPR 2024.